

1. a. Let $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$$p = \frac{a^T b}{a^T a} a$$

$$= \frac{[2 \ 1 \ 3] \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}{[2 \ 1 \ 3] \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$= \frac{2b_1 + b_2 + 3b_3}{14} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} (2b_1 + b_2 + 3b_3)/14 \\ (2b_1 + b_2 + 3b_3)/14 \\ (2b_1 + b_2 + 3b_3)/14 \end{bmatrix}$$

$$= \begin{bmatrix} 2/14 & 1/14 & 3/14 \\ 2/14 & 1/14 & 3/14 \\ 2/14 & 1/14 & 3/14 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

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~~A~~ Partial description of geometry
b. For better visualization:

$$P = \frac{1}{14} \begin{bmatrix} 2 & 1 & 3 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \end{bmatrix}$$

Column Space:

$$C(P) = \left\{ \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} \right\}$$

It is a line going through the origin and pointing to $(2, 2, 2)$ in $\mathbb{R}^3$.

Nullspace:

$$NCA = \left\{ \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix} \right\}$$

This is a plane going through the origin and points $(1, -2, 0)$ and $(0, -3, 1)$ in $\mathbb{R}^3$.

c. Since our matrix is singular, $\lambda_1 = 0$. And one more, the second $\lambda_2 = 0$. The remaining λ is the line of P .

$$\lambda_3 = 1/14 + 1/14 + 9/14 = 14/14 = 1$$

The diagonal entries of P add up to 1

The nullspace gives the perpendicular to line a . 2 evalues for $\lambda_1 = \lambda_2 = 0$
 $\lambda_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$, $\lambda_2 = \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix}$, and (A) gives $\lambda_3 = 1$'s evector $\lambda_3 = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$

2. $P = A \hat{x}$

a. If A has independent columns, this implies A must be invertible and so such:

$\hat{x} = (A^T A)^{-1} A^T b$. All the vectors perpendicular to $\hat{e} = b - A \hat{x}$ are the $C(A)$ vectors - the column of matrix A . If A is linearly dependent, A is no longer invertible and as such our formula is no longer correct.

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$$\begin{aligned} A &= QR \\ \hat{x} &= (A^T A)^{-1} A^T b \\ &= (Q R^T Q^T Q R)^{-1} (Q R^T)^T b \\ &= (R^T Q^T Q R)^{-1} R^T Q^T b \\ &= I R^T Q^T b \\ &= (R^T Q^T)^{-1} R^T Q^T b \\ P &= A \hat{x} \\ &= (QR)(R^T Q^T)^{-1} R^T Q^T b \\ &= Q R R^T Q^T b \\ &= Q Q^T b \end{aligned}$$

Should be same form

$$\begin{aligned} c. p &= [a_1 | a_2] \hat{x} \\ &= [a_1 | a_2] \left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \begin{bmatrix} a_1 & a_2 \end{bmatrix} \right)^{-1} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} b \\ &= [a_1 | a_2] (I)^{-1} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} b \\ &= [a_1 | a_2] \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} b \\ &= \begin{bmatrix} a_1^T b & a_2^T b \end{bmatrix} \end{aligned}$$

lol I didn't even check for orthogonality.

3. $A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ $A_3 = \begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & 0 \end{bmatrix}$

a. Using the provided methodology.

Least row: $[-(n-1) \ -(n-1) \ \dots \ -(n-1)]$
 $\Rightarrow -(n-1) [1 \ 1 \ \dots \ 1]$
 $\Rightarrow -(n-1) \det \begin{vmatrix} 0 & 0 & \dots & 0 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{vmatrix}$

$$\Rightarrow -(n-1) \det \begin{vmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{vmatrix}$$

$$\Rightarrow -(n-1)$$

$$A_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Ok method, using M_{11}

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With $n=3$:

$$A_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_3 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_4 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_5 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_6 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_7 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_8 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A_9 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Two evalues are: $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

$$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ and } x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

The trace is 0 but $\lambda_1 = \lambda_2 = 1$, therefore $\lambda_3 = -2$

A corresponding evector of $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ would be

$$x_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_5 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_6 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_7 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_8 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$